GATE CSE: 50 Important Questions (Functions, Arrays, and Recursion)

Functions (1–20)

What is the difference between Call by Value and Call by Reference?

Predict the output of a program involving nested function calls.

What is the effect of a static variable inside a function?

Determine the output of a recursive function.

Explain the use of function pointers in C.

Differentiate between scope and lifetime of a variable.

Trace the execution of multiple function calls.

What happens when a function returns a local variable's address?

How are parameters passed in C functions?

Explain pass-by-value with an example.

How many times is a function called in a given program?

Find the output of a function using global and local variables.

What is tail recursion?

Explain stack frame creation during function calls.

Predict the output involving static and automatic variables.

Explain recursive function execution.

Determine the order of execution of function calls.

What is the difference between library functions and user-defined functions?

Analyze the output of mutually recursive functions.

Write a function to compute GCD using recursion.

Arrays (21–40)

Explain one-dimensional arrays.

Explain two-dimensional arrays.

Find the output of an array traversal program.

Explain the relationship between arrays and pointers.

Predict the output of array initialization code.

How are strings represented as character arrays?

Find the largest element in an array.

Find the smallest element in an array.

Reverse an array.

Sort an array in ascending order.

Sort an array in descending order.

Explain binary search on arrays.

Explain linear search on arrays.

Analyze the time complexity of array traversal.

Explain multidimensional array memory layout.

Determine the address of an array element.

Explain pointer arithmetic with arrays.

What is an array of pointers?

What is a pointer to an array?

Pass an array to a function and explain what happens.

Recursion (41–50)

What is recursion?

Explain the concept of a base case.

Write a recursive program to find factorial.

Write a recursive program to generate Fibonacci numbers.

Find the output of a recursive function.

Determine the number of recursive calls made.

Analyze the stack space used by recursion.

Solve the recurrence relation $T(n) = T(n - 1) + 1$.

Solve the recurrence relation $T(n) = 2T(n/2) + n$.

Compare recursion and iteration.